~~1~~6. **SAT Scores** The SAT subject tests in chemistry and physics for two groups of 15 students each electing to take these tests are given here.

$s_1^2 / s_2^2 = 114^2 / 103^2$

$= 1.0178$

Chemistry	Physics
$\bar{x} = 784$	$\bar{x} = 758$
$s = 114$	$s = 103$
$n = 15$	$n = 15$

To use the two-sample t-test with a pooled estimate of σ^2, you must assume that the two population variances are equal. Test this assumption using the F-test for equality of variances. What is the approximate p-value for the test?

$H_0: \frac{\sigma_1^2}{\sigma_2^2} = 1 \quad H_a: \frac{\sigma_1^2}{\sigma_2^2} \neq 1$

$\alpha = 0.05$

$F_{STAT} = \frac{s_1^2}{s_2^2} \sim F_{n_1-1,\, n_2-1}$

$F^* = 1.225$

$F_{14,14,0.025}$ not available, use $F_{15,14,0.025} = 2.95$

RR: $F > 2.95$

$2P(F_{14,14} > 1.225) > 2P(F > 2.01) = 0.2 > 0.05$

$\Rightarrow$ no reject H_0